A brief (1-page) PDF document explaining your system architecture, the trade-offs you made, and how you addressed potential failure points.

Explanation on System Architecture

This system adopts a Retrieval-Augmented Generation (RAG) approach to improve response accuracy by retrieving information from the knowledge base before generate response through Large Language Model (LLM).

Retrieval-Augmented Generation (RAG)

RAG is selected because it allows the AI agent to generate responses based on the latest company information.

Pros	Cons
Easy to update knowledge base	Increase latency
Low development cost	Quality depends on documents' accuracy
Easy to maintain	

Large Language Model (LLM)

This system uses Large Language Model (LLM) for response generation because it provides better natural language understanding and produces more human-like responses. However, using LLM introduces several trade-offs such as using higher cost and potential hallucination where the model may generate inaccurate information if the context retrieved is insufficient.

Thus, to reduce these risks, the system combines LLM and RAG approach.

Human Escalation

Other than that, human escalation also included in order to handle sensitive and unresolved customer issues. The benefit is improved customer experience by ensuring critical cases are reviewed properly by human agents. On the other hand, this approach might increase operational cost due to the need of human resources.

Potential Failure Points and Mitigation

Potential failure might occur is when customer data leakage, thus the guardrail is implemented in order to check and prevent sensitive info being exposed. Other that that, there is a situation where AI unable to solve the issue, thus human escalation workflow is needed. Lastly, knowledge-based information might be outdated thus we need to regular review and update the documents in the system.